

```

#include <iostream>
using namespace std;

class People {
protected:
    int age;
    int height;
    int weight;
    static int num;

public:
    People(int a, int h, int w) {
        age = a;
        height = h;
        weight = w;
        num++;
    }

    //进食
    void Eating() {
        weight += 1;
    }

    //运动
    void Sporting() {
        height += 1;
    }

    //睡觉
    void Sleeping() {
        age += 1;
        height += 1;
        weight += 1;
    }

    //显示人的年龄、身高、体重
    void Show() {
        cout << "年龄： " << age << "岁
" << endl;
        cout << "身高： " << height <<
"厘米" << endl;
        cout << "体重： " << weight <<
"市斤" << endl;
    }
};

//显示人的个数
static void ShowNum() {
    cout << "当前人数： " << num
<< "人" << endl;
}

int People::num = 0;

int main() {
    People p1(20, 170, 120);
    People p2(25, 165, 110);

    cout << "初始状态" << endl;
    cout << "p1" << endl;
    p1.Show();
    cout << "p2" << endl;
    p2.Show();
    p1.ShowNum();

    cout << "\np1 进食后" << endl;
    p1.Eating();
    p1.Show();

    cout << "\np2 运动后" << endl;
    p2.Sporting();
    p2.Show();

    cout << "\np1 睡眠后" << endl;
    p1.Sleeping();
    p1.Show();

    cout << "\n 最终状态" << endl;
    cout << "p1" << endl;
    p1.Show();
    cout << "p2" << endl;
    p2.Show();
    p1.ShowNum();

    return 0;
}

```

```
初始状态
p1
年龄：20岁
身高：170厘米
体重：120市斤
p2
年龄：25岁
身高：165厘米
体重：110市斤
当前人数：2人
```

```
p1进食后
年龄：20岁
身高：170厘米
体重：121市斤
```

```
p2运动后
年龄：25岁
身高：166厘米
体重：110市斤
```

```
p1睡眠后
年龄：21岁
身高：171厘米
体重：122市斤
```

```
最终状态
p1
年龄：21岁
身高：171厘米
体重：122市斤
p2
年龄：25岁
身高：166厘米
体重：110市斤
当前人数：2人
```

```
#include <iostream>
#include <cstring>
using namespace std;

class Student {
private:
    char name[18];
    int num;
    int mathScore;
    int englishScore;
    static int count;
    static int mathTotalScore;
    static int englishTotalScore;

public:
    Student(const char* nm, int nu, int
math, int english) {
        strncpy(name, nm, 17);
        name[17] = '\0';
        num = nu;
        mathScore = math;
```

```
        englishScore = english;

        count++;
        mathTotalScore += math;
        englishTotalScore += english;
    }

    void ShowBase() {
        cout << "姓名： " << name <<
endl;
        cout << "学号： " << num <<
endl;
        cout << "数学成绩： " <<
mathScore << endl;
        cout << "英语成绩： " <<
englishScore << endl;
        cout << "\n" << endl;
    }

    static void showStatic() {
        cout << "统计信息" << endl;
        cout << "学生人数： " << count
<< endl;
        cout << "数学总成绩： " <<
mathTotalScore << endl;
        cout << "英语总成绩： " <<
englishTotalScore << endl;
        if (count > 0) {
            cout << "数学平均分： "
<< (double)mathTotalScore / count <<
endl;
            cout << "英语平均分： "
<< (double)englishTotalScore / count <<
endl;
        }
    }
};
```

```
int Student::count = 0;
int Student::mathTotalScore = 0;
int Student::englishTotalScore = 0;
```

```
int main() {
```

```

Student s1("周一", 2301, 77, 99);
Student s2("周二", 2302, 98, 88);
Student s3("周三", 2303, 42, 86);
Student s4("周四", 2304, 96, 95);

cout << "学生基本信息" << endl;
s1.ShowBase();
s2.ShowBase();
s3.ShowBase();
s4.ShowBase();
Student::showStatic();

return 0;
}

```

```

学生基本信息
姓名：周一
学号：2301
数学成绩：77
英语成绩：99

```

```

姓名：周二
学号：2302
数学成绩：98
英语成绩：88

```

```

姓名：周三
学号：2303
数学成绩：42
英语成绩：86

```

```

姓名：周四
学号：2304
数学成绩：96
英语成绩：95

```

```

统计信息
学生人数：4
数学总成绩：313
英语总成绩：368
数学平均分：78.25
英语平均分：92

```

```

#include <iostream>
#include <cstring>
using namespace std;

class Dog {
private:
    char* name;
    int age;
    char sex;
    double weight;

public:
    Dog() {
        name = new char[1];
        name[0] = '\0';
        age = 0;
        sex = 'M';
        weight = 0.0;
    }

    Dog(const char* n, int a, char s,
        double w) {
        name = new char[strlen(n) + 1];
        strcpy(name, n);
        age = a;
        sex = s;
        weight = w;
    }

    Dog(const Dog& other) {
        name = new
char[strlen(other.name) + 1];
        strcpy(name, other.name);
        age = other.age;
        sex = other.sex;
        weight = other.weight;
    }

    Dog& operator=(const Dog& other) {
        if (this != &other) {
            delete[] name;
            name = new
char[strlen(other.name) + 1];
            strcpy(name, other.name);

```

```

        age = other.age;
        sex = other.sex;
        weight = other.weight;
    }
    return *this;
}

~Dog() {
    delete[] name;
}

void setName(const char* n) {
    delete[] name;
    name = new char[strlen(n) + 1];
    strcpy(name, n);
}

const char* getName() const {
    return name;
}

void setAge(int a) {
    age = a;
}

int getAge() const {
    return age;
}

void setSex(char s) {
    sex = s;
}

char getSex() const {
    return sex;
}

void setWeight(double w) {
    weight = w;
}

double getWeight() const {
    return weight;
}

void display() const {
    cout << "姓名: " << name <<
endl;
    cout << "年龄: " << age << "岁
" << endl;
    cout << "性别: " << (sex == 'M' ?
"公": "母") << endl;
    cout << "体重: " << weight <<
"kg" << endl;
    cout << "\n" << endl;
}

void eat(double food) {
    weight += food * 0.1;
    cout << name << "吃了 " <<
food << "kg 食物, 体重增加了" << food
* 0.1 << "kg" << endl;
}

void run(double distance) {
    weight -= distance * 0.01;
    cout << name << "跑了 " <<
distance << "公里, 体重减少了" <<
distance * 0.01 << "kg" << endl;
}

};

int main() {

    Dog* dog1 = new Dog("小黄", 2, 'M',
10.0);
    Dog* dog2 = new Dog("小白", 1, 'F',
8.1);

    cout << "dog1" << endl;
    dog1->display();
    cout << "dog2" << endl;
    dog2->display();

    dog1->eat(0.5);
    dog1->run(2);
    dog1->display();
}

```

```

        delete dog1;
        delete dog2;

        return 0;
    }

```

```

dog1
姓名：小黄
年龄：2岁
性别：公
体重：10kg

dog2
姓名：小白
年龄：1岁
性别：母
体重：8.1kg

小黄吃了0.5kg食物，体重增加了0.05kg
小黄跑了2公里，体重减少了0.02kg
姓名：小黄
年龄：2岁
性别：公
体重：10.03kg

```

```

#include <iostream>
#include <string>
#include <iomanip>
#include <vector>
using namespace std;

class SavingsAccount {
private:
    string accountNumber;
    string accountName;
    string password;
    double balance;
    double annualInterestRate;

public:
    SavingsAccount(const string&
accNum, const string& accName,
const string& pwd,
double initialBalance, double interestRate
= 0.0035) {
        accountNumber = accNum;
        accountName = accName;
        password = pwd;

```

```

        balance = initialBalance;
        annualInterestRate =
interestRate;
    }

```

```

// 存款
bool deposit(double amount) {
    if (amount <= 0) {
        cout << "存款金额必须大
于 0！" << endl;
        return false;
    }
    balance += amount;
    cout << "存款成功！存入金额：
" << amount << "元" << endl;
    cout << "当前余额：" <<
balance << "元" << endl;
    return true;
}

```

```

// 取款
bool withdraw(double amount, const
string& inputPwd) {
    if (inputPwd != password) {
        cout << "密码错误，取款
失败！" << endl;
        return false;
    }
    if (amount <= 0) {
        cout << "取款金额必须大
于 0！" << endl;
        return false;
    }
    if (amount > balance) {
        cout << "余额不足！当前
余额：" << balance << "元" << endl;
        return false;
    }
    balance -= amount;
    cout << "取款成功！取出金额：
" << amount << "元" << endl;
    cout << "当前余额：" <<
balance << "元" << endl;
    return true;
}

```

```

    }

    // 计算年利息
    double    calculateAnnualInterest()
const {
    return    balance    *
annualInterestRate;
}

    // 计算月利息
    double    calculateMonthlyInterest()
const {
    return calculateAnnualInterest()
/ 12;
}

    // 设置年利率
    void setInterestRate(double newRate)
{
    if (newRate >= 0) {
        annualInterestRate    =
newRate;
        cout << "年利率已更新为：
" << newRate * 100 << "%" << endl;
    } else {
        cout << "利率不能为负数！
" << endl;
    }
}

    // 打印账户信息
    void printAccountInfo() const {
        cout << "\n 账户信息" << endl;
        cout << " 账 号 : " <<
accountNumber << endl;
        cout << " 户 名 : " <<
accountName << endl;
        cout << "余额: " << fixed <<
setprecision(2) << balance << "元" <<
endl;
        cout << " 年 利 率 : " <<
annualInterestRate * 100 << "%" << endl;

```

```

        cout << " 月 利 息 : " <<
calculateMonthlyInterest() << "元" <<
endl;
        cout << " 年 利 息 : " <<
calculateAnnualInterest() << "元" << endl;
        cout << "\n" << endl;
    }

    // 获取账号
    string getAccountNumber() const {
        return accountNumber;
    }

    // 获取户名
    string getAccountName() const {
        return accountName;
    }

    // 获取余额
    double getBalance() const {
        return balance;
    }

    // 获取年利率
    double getInterestRate() const {
        return annualInterestRate;
    }

    // 验证密码
    bool verifyPassword(const string&
inputPwd) const {
        return inputPwd == password;
    }
};

class Bank {
private:
    vector<SavingsAccount> accounts;

public:
    // 开户
    void openAccount() {
        string accNum, accName, pwd;
        double initialBalance;

```

```

        return &accounts[i];
    }
}

return 0;
}

// 存款
void deposit() {
    string accNum;
    double amount;

    cout << "\n 存款 " << endl;
    cout << "请输入账号： ";
    cin >> accNum;

    SavingsAccount* account =
    findAccount(accNum);
    if (account == 0) {
        cout << "账号不存在！ "
        << endl;
        return;
    }

    cout << "请输入存款金额： ";
    cin >> amount;
    account->deposit(amount);
}

// 取款
void withdraw() {
    string accNum, pwd;
    double amount;

    cout << "\n 取款" << endl;
    cout << "请输入账号： ";
    cin >> accNum;

    SavingsAccount* account =
    findAccount(accNum);
    if (account == 0) {
        cout << "账号不存在！ "
        << endl;
        return;
    }

    cout << "\n 开户" << endl;
    cout << "请输入账号： ";
    cin >> accNum;
    cout << "请输入户名： ";
    cin >> accName;
    cout << "请输入密码： ";
    cin >> pwd;
    cout << "请输入初始存款金额：
    ";
    cin >> initialBalance;

    // 检查账号是否已存在
    for (size_t i = 0; i <
    accounts.size(); i++) {
        if
        (accounts[i].getAccountNumber() ==
        accNum) {
            cout << "账号已存
            在，开户失败！ " << endl;
            return;
        }
    }

    if (initialBalance < 0) {
        cout << "初始存款不能为
        负数！ " << endl;
        return;
    }

    accounts.push_back(SavingsAccount(acc
    Num, accName, pwd, initialBalance));
    cout << "开户成功！ " << endl;
}

// 根据账号查找账户
SavingsAccount* findAccount(const
string& accNum) {
    for (size_t i = 0; i <
    accounts.size(); i++) {
        if
        (accounts[i].getAccountNumber() ==
        accNum) {

```

```

        cout << "请输入密码： ";
        cin >> pwd;
        cout << "请输入取款金额： ";
        cin >> amount;
        account->withdraw(amount,
pwd);
    }

    // 查询账户信息
    void queryAccount() {
        string accNum, pwd;

        cout << "\n 查询账户" << endl;
        cout << "请输入账号： ";
        cin >> accNum;

        SavingsAccount* account =
findAccount(accNum);
        if (account == 0) {
            cout << "账号不存在！ "
<< endl;

            return;
        }

        cout << "请输入密码： ";
        cin >> pwd;
        if
(account->verifyPassword(pwd)) {

account->printAccountInfo();
        } else {
            cout << "密码错误！ " <<
endl;
        }
    }
};

// 菜单函数
void displayMenu() {
    cout << "\n 个人活期账户管理系统
" << endl;
    cout << "1. 开户" << endl;
    cout << "2. 存款" << endl;

    cout << "3. 取款" << endl;
    cout << "4. 查询账户" << endl;
    cout << "0. 退出" << endl;
    cout << "请选择操作： ";
}

int main() {
    Bank bank;
    int choice;

    do {
        displayMenu();
        cin >> choice;

        switch (choice) {
            case 1:
                bank.openAccount();
                break;
            case 2:
                bank.deposit();
                break;
            case 3:
                bank.withdraw();
                break;
            case 4:
                bank.queryAccount();
                break;
            case 0:
                cout << "已退出！ "
<< endl;

                break;
            default:
                cout << "无效选择，
请重新输入！ " << endl;
        }
    } while (choice != 0);

    return 0;
}

```

个人活期账户管理系统

1. 开户
2. 存款
3. 取款
4. 查询账户
0. 退出

请选择操作：1

开户

请输入账号：aaa

请输入户名：bbb

请输入密码：abcdef

请输入初始存款金额：1000

开户成功！

个人活期账户管理系统

1. 开户
2. 存款
3. 取款
4. 查询账户
0. 退出

请选择操作：2

存款

请输入账号：aaa

请输入存款金额：2000

存款成功！存入金额：2000元

当前余额：3000元

个人活期账户管理系统

1. 开户
2. 存款
3. 取款
4. 查询账户
0. 退出

请选择操作：3

取款

请输入账号：aaa

请输入密码：abcdef

请输入取款金额：500

取款成功！取出金额：500元

当前余额：2500元

个人活期账户管理系统

1. 开户
2. 存款
3. 取款
4. 查询账户
0. 退出

请选择操作：4

查询账户

请输入账号：aaa

请输入密码：abcdef

账户信息

账号：aaa

户名：bbb

余额：2500.00元

年利率：0.35%

月利息：0.73元

年利息：8.75元